Ultimate Quantum Tic-Tac-Toe

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Abstract

We introduce Ultimate Quantum Tic-Tac-Toe, a quantum variant of the classic game of Tic-Tac-Toe which combines the superposition moves of Quantum Tic-Tac-Toe with the multi-board structure of Ultimate Tic-Tac-Toe. The rules given here define the game together with a stepping-stone named Quantum Tic-Tac-Toe Squared, and note where they modify or deviate from the published rules of the parent games.

1 Introduction

As its name indicates, *Ultimate Quantum* Tic-Tac-Toe is the offspring of two games, both described in [Orlin2022]: *Quantum* Tic-Tac-Toe and *Ultimate* Tic-Tac-Toe,

To understand its rules, it is helpful to first understand the rules of its parents, as well as to introduce an intermediate stepping-stone game which we have named Quantum Tic-Tac-Toe Squared.[1]

2 The parent games

Quantum Tic-Tac-Toe. This game was originally introduced in [Goff2006]. It is played on a 3×3 board, and adds a single superposition rule to the classic child's game. We present a slightly modified version here.

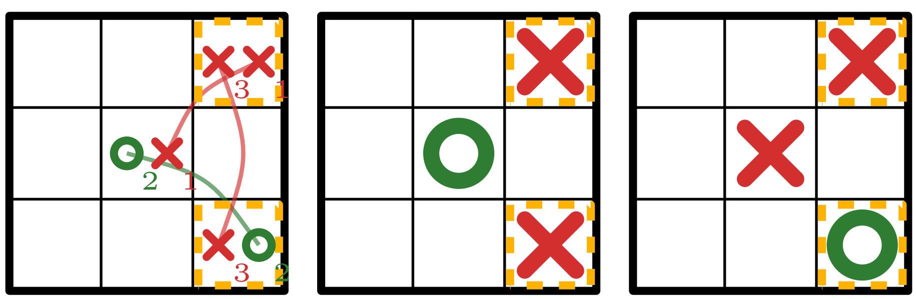
- All squares are initially open to play.
- X plays odd-numbered moves, O plays even-numbered moves.
- On each move, the current player marks two distinct open squares on the board which are not already connected[2] with their symbol, subscripted with the move number, and connects the symbols with a line.
- The two marks (and the squares where they reside) are entangled.

[1] All four games are playable at https://yonge-mallo.net/games/uqttt/.

[2] Both [Goff2006] and [Orlin2022] allow 2-cycles, but we disallow them in our version as experience showed that they limit the strategy in the offspring games. In particular, without this restriction, a player could always play the same pair twice on successive turns, forcing their classical marks onto both squares regardless of the opponent's actions.

- These marks are “spooky” and multiple “spooky” marks may be placed in the same square.
- Players take turns making moves until a cycle is formed, at which point a collapse is triggered which turns each pair of “spooky” marks into a classical symbol.
- Only one classical symbol is allowed per square, and a square with a classical symbol becomes closed to further moves.
- The player who did not cause the collapse chooses which of the two possible outcomes is realised for the cycle (and all stems hanging off of it, in a cascade). The board is redrawn with a classical symbol replacing one of the two “spooky” marks of each pair (and erasing the other “spooky” mark and the line between them).
- Three classical symbols in a line wins the game.
- If both players have three classical symbols in a line, it is a shared win. A board with 8 or 9 classical marks and no win is a draw.[3]

Board positions are numbered from 1 to 9 starting from the top left, reading left-to-right and then top-to-bottom. A move $p_1 - p_1$ entangles squares p_1 and p_2 (by convention $p_1 < p_2$ so moves are identified uniquely).



(a) (b) (c)

Figure 1: (a) A Quantum Tic-Tac-Toe game after the moves 3−5, 5−9, and 3−9. (b) One possible collapse outcome. (c) The other possible collapse outcome.

(Classical) Ultimate Tic-Tac-Toe. This game is played on a 3×3 grid of mini-boards, each of which is itself a 3×3 board.

The mini-boards are numbered from 1 to 9 in the analogous fashion to the board positions. We write b/p to refer to the p position of the b mini-board.

[3] The original [Goff2006] paper has a scoring system, tie-breaking rules in the case of a shared win, and places X in the final square if there are 8 classical marks, but we ignore these in our version.

For example, 2/6 refers to the right-most square in the middle row (position 6) of the middle top (position 2) mini-board.

- X begins the game by placing an X in any square on any mini-board.
- Players then take turns placing their symbol on the mini-board corresponding to the position of the previous move, within its mini-board. For example, if X plays 2/6, then O must play on mini-board 6. This rule is called the "sending rule".
- If that mini-board is already won or full, the next player has "free play" (i.e., they may play in any mini-board).[4]
- A mini-board is won by three symbols in a line, and the game is won by three won mini-boards in a line.

3 The stepping-stone game

Ultimate Quantum Tic-Tac-Toe is built from combining Quantum Tic-Tac-Toe with Ultimate Tic-Tac-Toe in two steps. The rules of the parent games carry over unchanged in the obvious manner in each step except where explicitly noted.

Quantum Tic-Tac-Toe Squared. This stepping-stone game extends Quantum Tic-Tac-Toe to a 3×3 board of mini-boards *without a sending rule*.

- Play proceeds as in Quantum Tic-Tac-Toe, except that players may entangle squares either within the same mini-board (intra-board entanglements) or across mini-boards (inter-board entanglements), subject to the rule that at most one entanglement exists between each pair of mini-boards.[5]
- Cycles and cascades work as previously, but may now span mini-boards. Open squares on won mini-boards remain open.[6] In particular, it is possible for a mini-board won by one player to be turned into a shared win later.
- Three won mini-boards in a line wins the game. A shared win on a mini-board counts for both players. If both players have three won mini-boards in a line, then the overall game is a shared win. If no legal moves remain and nobody has won, the game is a draw.

[4] In [Bertholon2020] it was shown that this rule is necessary for won mini-boards as otherwise the first player has an optimal strategy to force a win.

[5] This is a "lifted" version of the rule disallowing 2-cycles within each mini-board, and it has the same rationale. Experience also showed that allowing multiple entanglements between the same pair of mini-boards simply made the game too complex to understand, whereas this restriction limits the number of inter-board entanglements to $\binom{9}{2} = 36$.

[6] Note that this differs from the rule in Ultimate Tic-Tac-Toe, in which a won mini-board becomes closed to further play.

3.1 Ultimate Quantum Tic-Tac-Toe

Ultimate Quantum Tic-Tac-Toe is simply Quantum Tic-Tac-Toe Squared with a quantum version of the sending rule from classical Ultimate extended to entangled pairs.

- Play proceeds as in Squared, with the additional sending rule that the endpoint positions of the pair just played, $b_1/p_1 - b_2/p_2$, specify the next allowable move for the opponent. If $p_1 \neq p_2$, the opponent's next pair must entangle mini-boards p_1 and p_2; otherwise, the opponent's next pair must be an intra-board pair entirely in mini-board p_1 $(= p_2)$. For example, if X plays $2/6 - 9/4$, then O must entangle mini-boards 4 and 6; if O then plays $4/8 - 6/8$, X must follow by playing entirely within mini-board 8 on the next move.
- If there are no possible moves under the above constraint,[7] then the next player has "free play" (i.e., may play any pair which is legal under Squared).

Otherwise, every rule of Squared holds unchanged. In particular, open squares on won mini-boards remain open to play, and a player sent to a won mini-board must still play there, since open squares on a won mini-board may still matter to future collapses.[8]

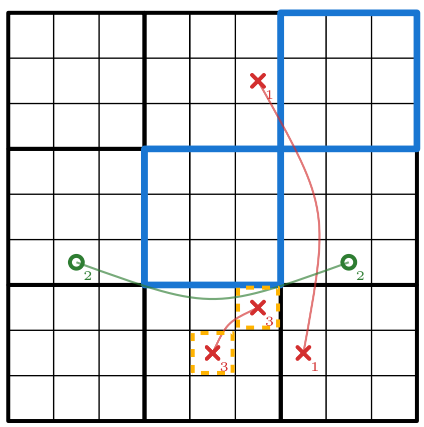


Figure 2: A game of Ultimate Quantum Tic-Tac-Toe in progress illustrating the sending rule. X's first move, $2/6 - 9/4$, sent O to mini-boards 4 and 6. O's move, $4/8 - 6/8$, then sent X to mini-board 8. X then played $8/3 - 8/5$, and so O's next move must entangle mini-boards 3 and 5.

[7] This can happen if: (1) the required inter-board entanglement is between two mini-boards which are already entangled; or (2) the required intra-board entanglement is on a mini-board with fewer than two open squares, or in which every pair of open squares is already entangled.

[8] The optimal strategy from [Bertholon2020] does not transfer, due to numerous differences between the two games.

References

[Orlin2022] Ben Orlin, *Math Games with Bad Drawings: 75 1/4 simple, challenging, go-anywhere games – and why they matter*, Black Dog & Leventhal, 2022.

[Goff2006] Allan Goff, *Quantum tic-tac-toe: A teaching metaphor for superposition in quantum mechanics*, American Journal of Physics, vol. 74, issue 11, pp. 962-973, Nov. 2006.

[Bertholon2020] Guillaume Bertholon, Rémi Géraud-Stewart, Axel Kugelmann, Théo Lenoir, and David Naccache, *At Most 43 Moves, At Least 29: Optimal Strategies and Bounds for Ultimate Tic-Tac-Toe*, arXiv:2006.02353, 2020.